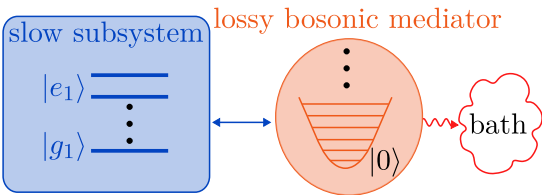


(a) Microscopic model



$$\hat{H} = \hat{H}_S - \Delta_c \hat{a}^\dagger \hat{a} + \hat{a}^\dagger \hat{F} + \hat{a} \hat{F}^\dagger,$$

$$\dot{\rho} = -i[\hat{H}, \rho] + \kappa \mathcal{D}[\hat{a}]\rho$$

$$\mathcal{D}[\hat{L}]\rho = \hat{L}\rho\hat{L}^\dagger - \frac{1}{2}\{\hat{L}^\dagger\hat{L}, \rho\}$$

(b) Effective resources

$$\chi_c(\omega) = \frac{1}{-(\Delta_c + \omega) - i\kappa/2}$$

induced Hamiltonian

$$\hat{H}_{\text{ind}} = \frac{\Delta_c}{\Delta_c^2 + \kappa^2/4} \hat{F}^\dagger \hat{F}$$

effective jump operator

$$\hat{L}_{\text{eff}} = \frac{\sqrt{\kappa}}{-\Delta_c - i\kappa/2} \hat{F}$$

near resonance

$$|\Delta_c| \lesssim \kappa$$

coherent dominant

far detuned

$$|\Delta_c| \gg \kappa$$

dissipative dominant

(c) Validity and breakdown

$$\eta_{\text{med}} = \frac{\|\hat{F}\|_{\mathcal{M}}}{\sqrt{\Delta_c^2 + \kappa^2/4}}$$



$$\eta_{\text{med}} \ll 1$$

controlled regime



$$\eta_{\text{med}} \sim 1$$

intermediate regime

memory effects emerging



$$\eta_{\text{med}} \gtrsim 1$$

breakdown

retain mediator or use memory-bearing model